Task 7 — Wireless Sensor Diagnostics

Protocol selection

- What we will do is compere all the 3 and then from the table we see what benifits us best in our case
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Criterion	Wi-Fi	Classic BT (SPP)	BLE
Latency	Low, but jitter is not so even basically	Low and steady	Low, but small packets and fixed intervals
Power	Highest, noisy bursts	Moderate	Lowest
Ease	Medium (needs AP, IP, client)	**Easiest**	Hardest (GATT setup)
Data rate vs. need	Far more than needed	**Enough** (~3 KB/s needed)	Enough, but must fragment data
Client	Browser or custom app	**Any serial terminal**	BLE app

- So what we will use is the WI-FI
    - **Latency**
        - Sensor glitches from bumps last only a few milliseconds, so I need data to arrive continuously and at a steady pace.
        - In Wifi mode, the debugger's phone or PC is the only device on the network. There is no shared traffic and no router in the path, so delay is low (roughly 5-20 ms) and steady.
        - We can apparently use a WebSocket so that as soon as the sensors get a reading we can ask the esp to push it to be on the web page ..
        - Classic Bluetooth SPP is similar in latency, so it has no real advantage here.
        - BLE sends small packets at fixed intervals, so data arrives in bursts and a short glitch can be smeared or missed.
    - **Power**
        - BLE uses the least power, but this is not a deciding factor. The motors draw far more than any BLE.
        - Wi-Fi draws the most radio current, in bursts of a few hundred mA. This is well within the 5 V rail budget (0.37 A maximum upar calculate kia he), and the motors run on their own 6 V buck, so the bursts will not cause a meaningful voltage drop i hbelieve. A capacitor near the ESP32's 5 V input should cover any small baxodi in voltage i believe.
        - So power ko dekh ke kisi cheez ko rule out nahi karenge coz almos saari hi similar he
    - **Ease of implementation**
        - The ESP32 hosts a small web page and a WebSocket server. Any phone or PC with a browser can connect, so no app has to be installed.
        - Writing the page (a live plot of the 8 sensors, with the faulty channel highlighted) is a small task, and the same page also provides the logging (a CSV download button).

- It works on Android, iOS, Windows and Linux alike. Classic Bluetooth SPP does not work on iPhones.
- Classic Bluetooth is simple for raw text, but it still needs a serial terminal app or a Python script to plot the data. That is about the same effort as the web page and gives a worse display.
- BLE needs GATT services and characteristics and a BLE app, which is the hardest of the three

- **4.Utkarsh Reasons**
  - Meri utkarsh me gaddi wifi baaar baar disconnect ho rhi thi .. ab baat ego pe aagyi he meri so issbar same problem aayigi toh me fix karke hi rahunga bc 😤😤🥀🥀

System architecture

